

Home Assignment_Deepeval Metrics

Home Assignment: DeepEval Metrics

Explore the different evaluation metrics available in the DeepEval GitHub repository.

Practise the following metrics using multiple examples:

1. Faithfulness
2. Answer Relevancy

For each metric, create and test different examples that demonstrate:

- A good response with a high score
- A partially correct response with a medium score
- A poor response with a low score

Observe the generated scores and reasons, and understand why each response received its respective score. Share your examples, execution results, and screenshots after completing the assignment.

Faithfulness

A good response with a high score

Vidhya's Workspace

Work locally with Git: Open your working directory in Postman to keep your API artifacts in sync with your codebase.

DeepEval_Metrics > Faithfulness Metric > Scenario 5- Good Faithfulness(Continuous Integration)

POST http://localhost:3000/api/eval-only

Body

```
1 {
2   "query": "What is Continuous Integration?",
3   "output": "Continuous Integration is the practice of frequently merging code changes and validating
4   them through automated builds and tests.",
5   "context": "Continuous Integration is a software development practice where developers regularly
6   integrate code changes into a shared repository, followed by automated build and testing processes.",
7   "metric": "faithfulness"
8 }
```

200 OK · 3.13 s · 981 B

Body

```
1 {
2   "evaluation": {
3     "metric": "faithfulness",
4     "score": 1,
5     "explanation": "The score is 1.00 because there are no contradictions, indicating perfect
6     faithfulness. Final score: 1.00.",
7     "results": [
8       {
9         "metric_name": "faithfulness",
10        "score": 1,
11        "explanation": "The score is 1.00 because there are no contradictions, indicating perfect
12        faithfulness. Final score: 1.00.",
13        "error": null
14      }
15    ]
16  }
```

Finish edits and lock request

Add negative-case assertions

Shape what's next

Tell us which features matter most to you. Takes under 1 minute.

Not now Take the survey

Vidhya's Workspace

Work locally with Git: Open your working directory in Postman to keep your API artifacts in sync with your codebase.

DeepEval_Metrics > Faithfulness Metric > Scenario 6- Good Faithfulness(Savings Account)

POST http://localhost:3000/api/eval-only

Body

```
1 {
2   "query": "What is a savings account?",
3   "output": "A savings account is a bank account that allows customers to deposit money and earn interest.",
4   "context": "A savings account is a bank account that allows customers to deposit money and earn
5   interest.",
6   "metric": "faithfulness"
7 }
```

200 OK · 3.34 s · 849 B

Body

```
1 {
2   "evaluation": {
3     "metric": "faithfulness",
4     "score": 1,
5     "explanation": "The score is 1.00 because there are no contradictions, indicating perfect
6     faithfulness. Final score: 1.00.",
7     "results": [
8       {
9         "metric_name": "faithfulness",
10        "score": 1,
11        "explanation": "The score is 1.00 because there are no contradictions, indicating perfect
12        faithfulness. Final score: 1.00.",
13        "error": null
14      }
15    ]
16  }
```

Add assertions for response body

Parameterize scenario fields

Shape what's next

Tell us which features matter most to you. Takes under 1 minute.

Not now Take the survey

A partially correct response with a medium score

The image displays two screenshots of the Postman application interface, showing API test results for two scenarios. Both screenshots are taken from a workspace named 'Vidhya's Workspace'.

Top Screenshot: Scenario 7-Partial Faithfulness(Generate Test cases)

- Request:** POST to `http://localhost:3000/api/eval-only`. The body is raw JSON:

```
{  "query": "Generate test cases for login.",  "context": [    "The application supports username and password authentication only."  ],  "output": "1. Verify login using username and password. 2. Verify biometric login. Biometric login is the only supported authentication method.",  "metric": "faithfulness"}
```
- Response:** 200 OK, 3.75 s, 1.11 KB. The response body is JSON:

```
{  "evaluation": {    "metric": "faithfulness",    "score": 0.3333333333333333,    "explanation": "The score is 0.33 because the actual output mentions biometric login support, which directly contradicts the retrieval context that states the application only supports username and password authentication. Final score: 0.33.",    "results": [      {        "metric_name": "faithfulness",        "score": 0.3333333333333333,        "explanation": "The score is 0.33 because the actual output mentions biometric login support, which directly contradicts the retrieval context that states the application only supports username and password authentication. Final score: 0.33."      }    ]  }
```

Bottom Screenshot: Scenario 8-Partial Faithfulness(fund transfer work)

- Request:** POST to `http://localhost:3000/api/eval-only`. The body is raw JSON:

```
{  "query": "How does fund transfer work?",  "context": [    "Users must complete OTP verification before transferring funds."  ],  "output": "Users must complete OTP verification before transferring funds. OTP verification is not required for fund transfers.",  "metric": "faithfulness"}
```
- Response:** 200 OK, 3.54 s, 1.13 KB. The response body is JSON:

```
{  "evaluation": {    "metric": "faithfulness",    "score": 0.5,    "explanation": "The score is 0.50 because the actual output incorrectly states that OTP verification is not required for fund transfers, directly contradicting the retrieval context which says users must complete OTP verification before transferring funds. Final score: 0.5.",    "results": [      {        "metric_name": "faithfulness",        "score": 0.5,        "explanation": "The score is 0.50 because the actual output incorrectly states that OTP verification is not required for fund transfers, directly contradicting the retrieval context which says users must complete OTP verification before transferring funds."      }    ]  }
```

Both screenshots show a sidebar with a list of scenarios and a 'New Chat' button on the right. A 'Shape what's next' pop-up is visible in the bottom right corner of both screenshots.

A poor response with a low score

The screenshot shows the Postman interface for a workspace named "Vidhya's Workspace". The active collection is "DeepEval_Metrics > Faithfulness Metric > Scenario 9-Bad Faithfulness(What is Playwright)". The request is a POST to "http://localhost:3000/api/eval-only". The response is a 200 OK with a body containing JSON data. The JSON data includes an "evaluation" object with "metric": "faithfulness", "score": 0, and an "explanation" that states the score is 0.00 because the actual output describes Playwright as a Microsoft browser automation framework, directly contradicting the retrieval context that defines Playwright as a stage performance for theatrical production. The response also includes a "results" array with a single result object.

```
1 {
2   "evaluation": {
3     "metric": "faithfulness",
4     "score": 0,
5     "explanation": "The score is 0.00 because the actual output describes Playwright as a Microsoft
6     browser automation framework, directly contradicting the retrieval context that defines
7     Playwright as a stage performance for theatrical production. Final score: 0.00.",
8     "results": [
9       {
10        "metric_name": "faithfulness",
11        "score": 0,
12        "explanation": "The score is 0.00 because the actual output describes Playwright as a
13        Microsoft browser automation framework, directly contradicting the retrieval context
14        that defines Playwright as a stage performance for theatrical production. Final score: 0.00."
15      }
16    ]
17  }
18 }
```

The screenshot shows the Postman interface for a workspace named "Vidhya's Workspace". The active collection is "DeepEval_Metrics > Faithfulness Metric > Scenario 10-Bad Faithfulness(What is Chrome)". The request is a POST to "http://localhost:3000/api/eval-only". The response is a 200 OK with a body containing JSON data. The JSON data includes an "evaluation" object with "metric": "faithfulness", "score": 0, and an "explanation" that states the score is 0.00 because the output incorrectly describes Google Chrome as a relational database management system, contradicting the context that it is a web browser developed by Google. The response also includes a "results" array with a single result object.

```
1 {
2   "evaluation": {
3     "metric": "faithfulness",
4     "score": 0,
5     "explanation": "The score is 0.00 because the output incorrectly describes Google Chrome as a
6     relational database management system, contradicting the context that it is a web browser
7     developed by Google. Final score: 0.00.",
8     "results": [
9       {
10        "metric_name": "faithfulness",
11        "score": 0,
12        "explanation": "The score is 0.00 because the output incorrectly describes Google Chrome
13        as a relational database management system, contradicting the context that it is a web
14        browser developed by Google. Final score: 0.00."
15      }
16    ]
17  }
18 }
```

Answer Relevancy

A good response with a high score

Vidhya's Workspace

Work locally with Git: Open your working directory in Postman to keep your API artifacts in sync with your codebase.

DeepEval_Metrics > Answer Relevancy Metric > Scenario 7-Good Relevancy (User Login)

POST http://localhost:3000/api/eval-only

Body

```
1 {
2   "query": "How do I verify that a user has logged in successfully?",
3   "output": "Verify that the user is redirected to the dashboard, the correct username is displayed,
4   and a valid authenticated session is created.",
5   "metric": "answer_relevancy"
6 }
```

200 OK · 2.52 s · 924 B

Body

```
1 {
2   "evaluation": {
3     "metric": "answer_relevancy",
4     "score": 1,
5     "explanation": "The score is 1.00 because the answer directly addressed how to verify a user's
6     successful login with no irrelevant content, earning a perfect relevancy rating.",
7     "results": [
8       {
9         "metric_name": "answer_relevancy",
10        "score": 1,
11        "explanation": "The score is 1.00 because the answer directly addressed how to verify a
12        user's successful login with no irrelevant content, earning a perfect relevancy rating."
13      }
14    ]
15  }
```

Fill the login request body

Add missing auth + env vars

Shape what's next

Tell us which features matter most to you. Takes under 1 minute.

Not now Take the survey

Vidhya's Workspace

Work locally with Git: Open your working directory in Postman to keep your API artifacts in sync with your codebase.

DeepEval_Metrics > Answer Relevancy Metric > Scenario 8-Good Relevancy (Invalid OTP)

POST http://localhost:3000/api/eval-only

Body

```
1 {
2   "query": "What should happen when a user enters an invalid OTP?",
3   "output": "The application should reject the OTP, display an appropriate error message, and allow
4   another attempt based on the configured retry limit.",
5   "metric": "answer_relevancy"
6 }
```

200 OK · 2.32 s · 934 B

Body

```
1 {
2   "evaluation": {
3     "metric": "answer_relevancy",
4     "score": 1,
5     "explanation": "The score is 1.00 because the answer directly addressed the question about handling
6     an invalid OTP with no irrelevant content, earning a perfect relevancy rating!",
7     "results": [
8       {
9         "metric_name": "answer_relevancy",
10        "score": 1,
11        "explanation": "The score is 1.00 because the answer directly addressed the question about
12        handling an invalid OTP with no irrelevant content, earning a perfect relevancy rating!"
13      }
14    ]
15  }
```

Create a negative-case body template

Add assertions for OTP failure

Shape what's next

Tell us which features matter most to you. Takes under 1 minute.

Not now Take the survey

A partially correct response with a medium score

The image displays two screenshots of the Postman interface, showing API test results for two scenarios. Both scenarios are part of a collection named 'DeepEval_Metrics' and are testing the 'Answer Relevancy Metric'.

Scenario 9-Partial Relevancy(Security Testing)

The request is a POST to `http://localhost:3000/api/eval-only`. The response is a 200 OK status, indicating a successful response. The response body is a JSON object:

```
{  "metric": "answer_relevancy",  "query": "What is security testing?",  "output": "Security testing identifies vulnerabilities and protects application data. Test cases can be stored in Jira, automation scripts can run in Chrome, and performance reports show response times."}
```

The test results show a score of 0.25. The explanation states: "The score is 0.25 because the response contained unrelated content about Jira test case storage, Chrome automation scripts, and performance reports, none of which address the question of what security testing is."

Scenario 10-Partial Relevancy(API Testing)

The request is a POST to `http://localhost:3000/api/eval-only`. The response is a 200 OK status, indicating a successful response. The response body is a JSON object:

```
{  "metric": "answer_relevancy",  "query": "What is API testing?",  "output": "API testing validates requests and responses between software systems. It verifies status codes and response data. It is also used to test the physical screen size of mobile devices."}
```

The test results show a score of 0.67. The explanation states: "The score is 0.67 because the response added an irrelevant comment about testing physical screen size of mobile devices, which isn't related to API testing, pulling the relevance down, though the core definition was still provided, keeping the score moderate."

A poor response with a low score

DeepEvalMetrics > Answer Relevancy Metric > Scenario 11- Bad Relevancy(Payment Transaction)

POST http://localhost:3000/api/eval-only

Body

```
1 {
2   "query": "What should happen when a payment transaction fails?",
3   "output": "A test plan describes the scope, objectives, resources, and schedule of testing activities.",
4   "metric": "answer_relevancy"
5 }
```

200 OK · 2.78 s · 837 B

Body

```
1 {
2   "evaluation": {
3     "metric": "answer_relevancy",
4     "score": 0,
5     "explanation": "The score is 0.00 because the output only described a test plan, which is unrelated to how a failed payment transaction should be handled.",
6     "results": [
7       {
8         "metric_name": "answer_relevancy",
9         "score": 0,
10        "explanation": "The score is 0.00 because the output only described a test plan, which is unrelated to how a failed payment transaction should be handled.",
11        "error": null
12      }
13    ]
14   }
15 }
```

Bring renamed request in sync
Add a negative test assertion

Shape what's next
Tell us which features matter most to you. Takes under 1 minute.
Not now Take the survey

DeepEvalMetrics > Answer Relevancy Metric > Scenario 12- Bad Relevancy(Email Notification)

POST http://localhost:3000/api/eval-only

Body

```
1 {
2   "query": "How can I verify that an email notification was sent successfully?",
3   "output": "Load testing evaluates application behaviour under a large number of concurrent users.",
4   "metric": "answer_relevancy"
5 }
```

200 OK · 2.66 s · 888 B

Body

```
1 {
2   "evaluation": {
3     "metric": "answer_relevancy",
4     "score": 0,
5     "explanation": "The score is 0.00 because the response focused on load testing, which is unrelated to verifying email notifications, so it did not address the question at all.",
6     "results": [
7       {
8         "metric_name": "answer_relevancy",
9         "score": 0,
10        "explanation": "The score is 0.00 because the response focused on load testing, which is unrelated to verifying email notifications, so it did not address the question at all.",
11        "error": null
12      }
13    ]
14   }
15 }
```

Turn this into a parameterized run
Add a second bad-relevancy case

Shape what's next
Tell us which features matter most to you. Takes under 1 minute.
Not now Take the survey